

Results1

07 March 2026 10:30

```
C:\Users\Santhosh147\AppData\Local\Programs\Python\
Python314\python.exe C:\Users\Santhosh147\Downloads\
mlproject\main.py
```

	Year	Quarter	...	Freight From Origin	Frieght To Origin
0	2025	1	...	967.99	320.05
1	2025	2	...	1162.54	343.54
2	2025	3	...	1661.87	1027.75
3	2025	3	...	82.31	54.96
4	2025	1	...	101.11	3.21

[5 rows x 8 columns]

	Year	Quarter	...	Freight From Origin
Frieght To Origin				
count	71.0	71.000000	...	71.000000
		71.000000		
mean	2025.0	2.028169	...	726.255493
		273.434507		
std	0.0	0.827593	...	1009.986845
		387.785073		
min	2025.0	1.000000	...	0.000000
		0.000000		
25%	2025.0	1.000000	...	54.525000
		3.300000		
50%	2025.0	2.000000	...	198.110000
		38.970000		
75%	2025.0	3.000000	...	938.955000
		448.275000		
max	2025.0	3.000000	...	3754.380000
		1484.410000		

[8 rows x 6 columns]

Year	0
Quarter	0
Origin	0
Dest	0

```

Pax From Origin      0
Pax To Origin        0
Freight From Origin  0
Frieght To Origin    0
dtype: int64
Origin
HYDERABAD      71
Name: count, dtype: int64
Dest
ABUDHABI      3
BAHRAIN       3
BANGKOK       3
COLOMBO       3
DAMMAM        3
DHAKA         3
DOHA          3
DUBAI         3
FRANKFURT     3
HONG KONG     3
ISTANBUL      3
JEDDAH        3
KUALALUMPUR   3
KUWAIT        3
LONDON        3
MEDINA        3
MUSCAT        3
PHUKET        3
RAS AL-KHAIMAH 3
RIYADH        3
SHARJAH       3
SINGAPORE     3
HO CHI MINH CITY 2
MALE          2
AMSTERDAM     1
Name: count, dtype: int64
Quarter
3      25
1      23
2      23
Name: count, dtype: int64
C:\Users\Santhosh147\AppData\Local\Programs\Python\

```

```

Python314\Lib\site-packages\numpy\lib\
_function_base_impl.py:3065: RuntimeWarning: invalid
value encountered in divide
  c /= stddev[:, None]
C:\Users\Santhosh147\AppData\Local\Programs\Python\
Python314\Lib\site-packages\numpy\lib\
_function_base_impl.py:3066: RuntimeWarning: invalid
value encountered in divide
  c /= stddev[None, :]
Selected Numeric Features: ['Pax To Origin', 'Freight
From Origin', 'Frieght To Origin']
Selected Categorical Features: []
Final Selected Features: ['Pax To Origin', 'Freight
From Origin', 'Frieght To Origin']
Validation MAE: 2841.526574132778
Validation MSE: 18745499.481791187
Validation R2 Score: 0.9846066740263838
Best Hyperparameters: {'regressor__alpha': 100}
Test MAE: 3538.9136932228807
Test MSE: 25790496.050472874
Test R2 Score: 0.9753392616345842
[LightGBM] [Info] Auto-choosing row-wise multi-
threading, the overhead of testing was 0.000516 seconds
.
You can set `force_row_wise=true` to remove the
overhead.
And if memory is not enough, you can set `
force_col_wise=true`.
[LightGBM] [Info] Total Bins 43
[LightGBM] [Info] Number of data points in the train
set: 42, number of used features: 3
[LightGBM] [Info] Start training from score 26747.
357143
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf

```

[illegible]

[illegible]

[illegible]

[illegible]


```
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive
gain, best gain: -inf
C:\Users\Santhosh147\AppData\Local\Programs\Python\
Python314\Lib\site-packages\sklearn\utils\validation.py
:2691: UserWarning: X does not have valid feature names
, but LGBMRegressor was fitted with feature names
  warnings.warn(
[('Linear Regression', 0.9753388951099251), ('Decision
Tree', 0.9599117085106476), ('Random Forest', 0.
9004216087666848), ('SVR', -0.07349118259465603), ('
Gradient Boosting', 0.9653495292477612), ('XGBoost', 0.
6349349021911621), ('LightGBM', 0.3479695287981277)]
Model saved as model.joblib

Process finished with exit code 0
```